

SalahAbdo at Daleel 2026: Routing Each Label to LLM Votes, a Fine-Tuned Encoder, or Sequence Structure

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Abstract

We describe the SalahAbdo closed-track systems for both Daleel 2026 subtasks: deciding which argumentative-unit types appear in an Arabic paragraph (Task 1) and locating their exact spans (Task 2), in editorials and debates. No single method was best for all six argumentative-unit types, so our Task 1 system routes each type to the method that measured best for it: assumption (AS), testimony (TE), and statistics (ST) are detected by sampling a gemma-4-31b classifier five times and firing when enough samples agree; anecdote (AN) and other (OT) by a fine-tuned CAMELBERT classifier; and the lowest-agreement type, common ground (CO), by ranking paragraphs and firing a fixed count. A final verifier shows the LLM examples of true and false ST paragraphs and removes likely false positives. Task 2 first extracts spans as verbatim LLM quotes, then re-labels them by blending encoder probabilities with label-sequence statistics under Viterbi decoding, with zero additional LLM calls. Every design change had to clear an adoption bar fixed before any experiment ran: 29 recorded decisions produced six adoptions, and composition arithmetic predicted two development scores digit-for-digit. Development scores are 0.7089 macro- F_1 (Task 1) and 0.7247 span F_1 (Task 2); the pinned recipes scored 0.6499 and 0.7055 on test, and our final official submissions scored **0.712** and **0.7316: first of ten and fifth of eight** closed-track teams (§6 discloses the evaluation-phase protocol). Residual errors track the scheme’s measured human agreement (κ as low as 0.114): types whose definitions a model can check against the text improved under LLM verification, while types defined by annotation convention resisted every prompting attack and were learned only from supervision and sequence structure.

1 Introduction

Argument mining is “the automatic identification and extraction of the structure of inference and reasoning expressed as arguments presented in natural language” (Lawrence and Reed, 2019). Daleel 2026 (Nabhani et al., 2026) asks for its first two steps (Peldszus and Stede, 2013) in Arabic: find the argumentative discourse units (ADUs) in a paragraph and decide which of six types each one is, in two genres. Prior Arabic resources target stance detection (Alhindi et al., 2021; Alturayef et al., 2022) and debate component classification (Khader et al., 2024; Al-Zawqari et al., 2025), not this six-type evidence-unit scheme. The task is hard in a measurable way: whether a clause is *common ground* or an *assumption* is implicit content that, as Moens (2018) observes (quoted in the survey), “resides in the mind of communicator and audience.” Even trained annotators agreed only slightly on such labels ($\kappa=0.114$; §2).

Criteria can be checked; conventions cannot.

Some Daleel labels are defined by criteria a model can *check* against the text: a statistics (ST) unit contains numbers or study results, a testimony (TE) unit attributes a statement to a source. Others are defined by conventions with no text-checkable rule to appeal to: whether shared background is common ground (CO) rather than an assumption (AS), or a settled event an anecdote (AN) rather than CO, is a line the annotation team drew. The distinction predicted our results: every LLM verification stage aimed at a checkable label worked, every prompting attack on a convention label failed, and convention labels entered the system only through a fine-tuned encoder and sequence structure.

Every change had to clear a bar set in advance.

A change was adopted only if it beat a score bar fixed before its results existed, with the design and

both planned outcomes recorded in a ledger; failed changes were closed for good (§5; all twenty-nine in Appendix B). This caught two changes that looked good internally but failed externally, and predicted two development scores digit-for-digit (§6).

Contributions. (1) A Task 1 system that routes each label to the method that measured best for it, plus a verifier for the one label whose problem was precision (dev macro- F_1 0.6159 $\rightarrow$ 0.7089). (2) A Task 2 re-labeling stage that improves an LLM span pipeline by +0.0400 dev F_1 with no additional LLM calls, plus a placebo ablation separating structure’s contribution from the encoder’s. (3) A gated adoption methodology for shared-task conditions, publishing the gate registry, negative results, transfer ledger, and an analysis tying residual errors to measured human agreement and vote statistics. Code, prompts, scorer ports, and analysis artifacts: <https://github.com/Salah-Sal/daleel-2026-arabic-argumentative-discourse-mining-shared-task> (public at camera-ready).

2 Task, Data, and the Human Ceiling

Setup. Daleel 2026 (Nabhani et al., 2026) provides Arabic paragraphs from two genres: news editorials and transcribed competitive debates (the Munazarat lineage; Khader et al., 2024; Al-Sharafi et al., 2025). Task 1 asks for the *set* of ADU types present in a paragraph (six labels, multi-label, empty sets legal); Task 2 asks for exact typed character spans. Example (editorial, translated): “Apple says privacy lies at the core of all its products, yet these products come at elevated prices” $\rightarrow$ Task 1 {TE, CO}; Task 2 a TE span over the Apple claim and a CO span over the pricing clause. Task 1 is scored by macro- F_1 over all six labels; Task 2 by a partial-match span F_1 in which each overlapping same-label (prediction, gold) pair adds overlap/len(prediction) to precision and overlap/len(gold) to recall (scorer ports parity-tested; Appendix D). The training release contains 612 paragraphs (357 debate, 255 editorial; 2,975 gold spans covering 91% of all text); the development input contains 217 and the test input 213. Label priors are heavily skewed: AS appears in 75% of paragraphs, other (OT) 36%, AN 30%, TE 26%, CO 6%, ST 5%. For all 612 paragraphs the Task 1 label set equals the Task 2 span-label set, so one span model serves both

tasks. We split the 612 training paragraphs into a 430-paragraph working pool (five genre-balanced folds; Appendix D) and a 182-paragraph held-out check. *Out-of-fold* (OOF) scoring fits on four folds and predicts the fifth; all design decisions used OOF scores on the 430, the 182 serving only as a sanity guard (§5).

The scheme and its measured ceiling. The six types, common ground (CO), assumption (AS), testimony (TE), statistics (ST), anecdote (AN), other (OT), are those of the Webis-Editorials-16 annotation model (Al-Khatib et al., 2016), whose original annotation study *measured* per-type agreement (Fleiss’ κ at segment level, before annotator judgments were merged into gold labels): from 0.613 for AS down to **0.114** for CO — slight agreement (Landis and Koch, 1977; Fleiss, 1971); full table in §C.1. Given one annotator chose CO, the probability another chose AS is 0.562, and CO/OT are also the rarest types in the source corpus (1.7%/1.2% of 14,313 units): low agreement and extreme rarity together bound what any system can learn. We extrapolate a pooled Task 2 ceiling of ≈ 0.75 (our estimate, not a published number) and read residual errors against it.

3 Task 1: Routed Multi-Source Composition

No single method was best for all six labels (§6), so our final Task 1 system (**v2.2**) *routes* each label to the source that measured best for it, with every threshold fitted on training folds and applied to held-out folds (Table 1).

Sources. (a) *Self-consistency votes*: the frozen zero-shot chain-of-thought classifier (gemma-4-31b-it, winner of a pre-registered 14-model bake-off) is sampled $k=5$ times at $T=0.7$ (Wang et al., 2023); a label fires when at least θ_L of the five samples propose it ($k=5$ suffices; §C.2). (b) *Encoder*: CAMEL-BERT-MSA-quarter (Inoue et al., 2021) fine-tuned to output a probability per label per paragraph (Appendix D), with decision thresholds fitted on OOF probabilities; the quarter-size variant was chosen in advance on external evidence (Al-Zawqari et al., 2025). (c) *Span-derived labels*: the Task 2 pipeline’s label sets, used only in the AN union rule and the CO ranker.

The ST verifier. An OOF error audit showed ST recall already at 0.947 (18 of 19 golds found)

Label	Source	Decision rule (deployment twin identical)	OOF F_1	Dev F_1
AS	LLM votes	fire iff $\geq \theta_{AS}=2$ of 5 rollout votes	0.9192	0.9046
TE	LLM votes	fire iff $\geq \theta_{TE}=4$ of 5	0.7782	0.8256
ST	votes + verifier	fire iff $\geq \theta_{ST}=4$ of 5, then drop iff verifier says absent <i>and</i> a second-family model omits ST	0.7826	0.8421
AN	encoder $\cup$ spans	encoder OOF decision $\cup$ (span-derived AN $\wedge$ votes ≥ 1)	0.7243	0.6309
OT	encoder	encoder OOF decision, as-is	0.7744	0.7344
CO	budgeted rank	top $2 \times$ prior per fold by rank-mean(encoder sigmoid, vote fraction, span presence)	0.4000	0.3158
macro			0.7298	0.7089

Table 1: The v2.2 Task 1 router. “Votes” = $k=5$, $T=0.7$ rollouts of the frozen zero-shot gemma-4-31b classifier; encoder = CAMELBERT-MSA-quarter (5-fold OOF); “spans” = Task 2 label sets; Dev = the final submission’s development scores.

while 19 of 37 fired paragraphs were false positives: a pure precision problem. The verifier shows the LLM what true and false ST paragraphs look like: for a fired paragraph, its prompt holds the official definition plus all gold-ST paragraphs and all fired false positives from the *other* folds (~ 15 each) as labeled examples; three samples at $T=0.7$ vote. A paragraph is dropped only if the verifier majority says absent *and* an independent model family (qwen3-32b or llama-3.3-70b) also omits ST — a two-key rule. On OOF it dropped 10 paragraphs, all false positives (ST F_1 0.6429 $\rightarrow$ 0.7826); on dev it dropped 2, again both false positives (0.7619 $\rightarrow$ 0.8421). The question is checkable, so verification transfers.

The CO budget. Nothing we tried could actually detect CO (§C.1), and for a label this uncertain, over-predicting is exactly what macro- F_1 rewards (Lipton et al., 2014). So we control how *much* fires rather than *what*: paragraphs are ranked by the mean rank of three signals (encoder probability, vote fraction, span presence), and the top $2 \times$ training-frequency count is fired — a count, unlike a threshold, stays stable when scores drift across datasets (cf. Lin and Lin, 2023).

4 Task 2: Span Pipeline + Structural Re-Decode

v3: proposal $\rightarrow$ atom $\rightarrow$ role. LLMs are unreliable at character arithmetic, so the model never outputs offsets: the extraction stage asks for (label, *verbatim quote*) pairs, and a staged aligner locates each quote in the paragraph (exact, whitespace-tolerant, orthography-normalized, then fuzzy; re-aligning gold spans’ own text scores F_1 0.999). The extracted text is then split at clause boundaries and connectives, and a classifier (SPANROLEDECISION) labels each piece with one of the six types or NONE; its prompt is the campaign’s only adopted prompt-optimizer product (GEPA (Agrawal et al., 2026) under DSPy

(Khattab et al., 2024); prompt selection never saw the scored fold). Extraction misses little: at most 4.2% of any label’s gold mass falls outside every predicted span, so residual Task 2 error is overwhelmingly mislabeling of correctly extracted text.

S1: the structural re-decode. v3 labels each extracted span *independently*, yet gold labels are strongly sequential: an AN span follows another AN 39% of the time, against an 11.5% base rate. S1 therefore re-labels each paragraph’s span sequence in five frozen steps: (1) a fine-tuned CAMELBERT segment classifier gives each span a probability distribution over the six labels; (2) each label’s probability is rescaled by the ratio (gold share of text) / (expected share under the encoder) — the $\tau=1$ point of logit adjustment (Menon et al., 2021); (3) each span’s evidence is $p = (1-\lambda)p_{\text{enc}} + \lambda \text{onehot}(\text{v3 label})$, $\lambda=0.9$: the v3 label keeps most of the weight; (4) Viterbi decoding (Viterbi, 1967) picks the most probable label sequence under genre-specific transition matrices estimated on training folds (Laplace $\alpha=1$); (5) Rule-A: a decoded CO span must also appear in an independent zero-shot run, else it takes that run’s dominant label. Everything is fitted per fold, and the bundle changes only labels, never boundaries, so segmentation credit is preserved by construction (Widmoser et al., 2021).

What S1 fixed. OOF pooled F_1 moved 0.6934 $\rightarrow$ 0.7206 (+0.0272, better on all 5 folds), changing 237 of 3,824 span labels. A *placebo ablation* — the identical decode with the encoder replaced by uniform distributions — scores 0.7133: structure alone contributes $\sim +0.02$, encoder evidence the remaining +0.0073 (cf. the $\neg$ DAPT control of Gururangan et al., 2020). What the decode repairs is over-prediction of rare labels: predicted CO text fell from $3.98 \times$ the gold amount to $1.78 \times$ and ST from $1.38 \times$ to $1.07 \times$. The declared limit: pre-

dicted AN text sits at $0.48\times$ gold before and $0.49\times$ after: most gold AN text lies inside larger spans of another label, which re-labeling cannot split.

5 Adoption Discipline

Register, then run. The adoption bar entered version control before any experiment ran; each candidate change was then written into the ledger: design, grid, baseline, and both planned outcomes. Where the entry’s commit precedes its run — both Task 2 adoptions among them — git order itself proves the plan came first; elsewhere design and verdict entered git together and precedence rests on the ledger (marked in Appendix B). This is *pre-registration*: committing to the plan before the results exist (van Miltenburg et al., 2021). (Ten early decisions predate the ledger.)

The gate. A change is adopted only if the composed system’s OOF score improves by at least $+0.02$ on the official metric *and* wins on at least 3 of the 5 folds. The margin tracks measured noise (identical LLM configurations rerun at ± 0.03 – 0.04 ; estimated composed-OOF band $\sigma \approx 0.015$); smaller gains are indistinguishable from it. The fold rule blocks adopting a spike on one lucky fold (CO has 25 positives, ST 19). Every gate run recomputes its baseline from identical inputs and aborts on mismatch; a rejected family closes permanently: 29 decisions, six adoptions, zero re-opened families.

One bundled probe; external checks are sacred. Each adopted system earned exactly one development-set submission; nothing was tuned against the leaderboard. Two failures justify it: an encoder swap that won a 4-way internal comparison collapsed on dev ($0.6955 \rightarrow 0.6364$), and a recomposed OT source reversed (-0.10 on its label); both were reverted. The 182-paragraph check proved unreliable too (its champion readout missed dev by -0.087), so composed OOF on the 430 became the honest predictor.

Exact composition arithmetic. Macro- F_1 averages six per-label scores and the router decides each label independently, so swapping one label’s source moves the system score by an exactly computable amount. This let us *predict* development scores: v2.1 and v2.2 were predicted 0.6955 and 0.7089, and scored exactly that (Table 5).

6 Results

Tables 5 and 6 give the transfer ledger and progression; adopted changes transferred, S1’s dev gain ($+0.0400$) *exceeding* its OOF gain ($+0.0272$). On test (213 paragraphs) the pinned recipes scored 0.6499 and 0.7055, the Task 2 move from dev (-0.019) inside the noise band. The Task 1 drop (-0.059) decomposes by label, not by system: AS, ST and OT improved on test, while the loss concentrates where frozen operating points met a shifted test set (CO, TE, AN); Appendix C decomposes both.

Evaluation-phase protocol (post-freeze, disclosed). When the organizers released the development gold labels and extended the deadline, we waived our own freeze for that window and ran one time-boxed, leaderboard-informed campaign: four probes per task in one evening plus two closing Task 2 uploads. Per-label scorer feedback moved the AS, TE, AN, and CO sourcing decisions across probes (Task 1 $0.6499 \rightarrow 0.712$; Task 2 $0.7055 \rightarrow 0.7316$, via span-drop filters keyed to Task 1 labels and encoder probabilities). Two stronger configurations missed the board through operational error (Appendix A). In the organizers’ final standings we place **first of ten** closed-track teams on Task 1 and **fifth of eight** on Task 2 (leader 0.755). Our Task 1 lead is 3×10^{-5} , far inside our own noise band: we read the top two as tied. We report these gains separately: our methodological claims rest on the gated development phase, not the probe campaign.

Analysis. Appendix C develops it: prompting fails on convention labels while verification transfers on checkable ones (§C.1); $k=5$ votes suffice and rare-label operating points sit on needles (§C.2); S1’s residual mirrors the human confusion matrix (§C.3). Paid API spend was $\approx$ US\$60 (Dodge et al., 2019).

7 Conclusion

Both systems place every component where measured evidence routed it, and every adoption cleared a pre-declared gate; the ranks matter less than the ledger behind them. At this scheme’s annotation ceiling *verification* helps where definitions are checkable, while convention labels move only under supervision, priors, and structure. The registry guards against the winner’s curse; extraction is the unreachable lever.

Limitations

All selection statistics rest on a 430-paragraph OOF pool with 25 CO and 19 ST positives; our estimated composed-OOF noise band is $\sigma \approx 0.015$, and our +0.02 gate is calibrated to it, not to a universal standard. The ≈ 0.75 Task 2 ceiling is our extrapolation from the source corpus’s agreement study (a different language, granularity, and consolidation stage) and serves as an interpretive anchor, not a bound; the comparison with the human confusion matrix in §C.3 sets differently normalized quantities side by side, so it is directional, not exact. LLM legs depend on a specific serving stack: the same checkpoint scored 0.048 lower on a different provider and identical configurations rerun at ± 0.03 –0.04; pinning the provider, caching artifacts, and count-based rare-label rules reduce but do not eliminate this risk. The ST verifier’s exemplar block is label-sorted, which long-context evidence suggests is suboptimal (Bertsch et al., 2024); the recipe was devalidated as-is. The TAPT (task-adaptive pretraining) measurement is single-seed. Our systems are tuned to this scheme’s conventions by design; we make no claim that the specific routing transfers to other argument-annotation schemes (“each data set appears to conceptualize claims quite differently,” Daxenberger et al., 2017; Lawrence and Reed, 2019). Finally, the evaluation-phase probe campaign was leaderboard-informed (§6); we report it separately and do not claim its gains as evidence for the methodology.

References

- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2026. [GEPA: Reflective prompt evolution can outperform reinforcement learning](#). In *The Fourteenth International Conference on Learning Representations (ICLR)*.
- Khalid Al-Khatib, Henning Wachsmuth, Johannes Kiesel, Matthias Hagen, and Benno Stein. 2016. [A news editorial corpus for mining argumentation strategies](#). In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 3433–3443, Osaka, Japan. The COLING 2016 Organizing Committee.
- Abdul Gabbar Al-Sharafi, Mohammad Majed Khader, Mohamad Hamza Al-Sioufy, Mohamed Ahmed, and Ali Al-Zawqari. 2025. Arabic argumentation: Presentation or proof. *Informal Logic*, 45(2):194–228.
- Ali Al-Zawqari, Mohamed Ahmed, Abdul Gabbar Al-Sharafi, Mohammad Majed Khader, Ali Safa, and Gerd Vandersteen. 2025. [Neural classification of argument elements and styles in Arabic competitive debates](#). *IEEE Access*, 13:115944–115959.
- Tariq Alhindi, Amal Alabdulkarim, Ali Alshehri, Muhammad Abdul-Mageed, and Preslav Nakov. 2021. [AraStance: A multi-country and multi-domain dataset of Arabic stance detection for fact checking](#). In *Proceedings of the Fourth Workshop on NLP for Internet Freedom: Censorship, Disinformation, and Propaganda (NLP4IF)*. Association for Computational Linguistics.
- Nora Saleh Alturayef, Hamzah Abdullah Luqman, and Moataz Aly Kamaleldin Ahmed. 2022. [Mawqif: A multi-label Arabic dataset for target-specific stance detection](#). In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 174–184. Association for Computational Linguistics.
- Amanda Bertsch, Maor Ivgi, Emily Xiao, Uri Alon, Jonathan Berant, Matthew R. Gormley, and Graham Neubig. 2024. [In-context learning with long-context models: An in-depth exploration](#). *Preprint*, arXiv:2405.00200.
- Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. 2024. [Are more LM calls all you need? towards the scaling properties of compound AI systems](#). *Preprint*, arXiv:2403.02419. Title as in arXiv v2; earlier version titled “Are More LLM Calls All You Need?”.
- Johannes Daxenberger, Steffen Eger, Ivan Habernal, Christian Stab, and Iryna Gurevych. 2017. [What is the essence of a claim? cross-domain claim identification](#). In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 2055–2066, Copenhagen, Denmark. Association for Computational Linguistics.
- Flavio Di Palo, Prateek Singhi, and Bilal Fadlallah. 2024. [Performance-guided LLM knowledge distillation for efficient text classification at scale](#). *Preprint*, arXiv:2411.05045.
- Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. 2019. [Show your work: Improved reporting of experimental results](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2185–2194, Hong Kong, China. Association for Computational Linguistics.
- Joseph L. Fleiss. 1971. [Measuring nominal scale agreement among many raters](#). *Psychological Bulletin*, 76(5):378–382.

- Jinlan Fu, Xuanjing Huang, and Pengfei Liu. 2021. [SpanNER: Named entity re-/recognition as span prediction](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics.
- Gemma Team, Google DeepMind. 2026. [Gemma 4 technical report](#). *Preprint*, arXiv:2607.02770.
- Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. [Don’t stop pretraining: Adapt language models to domains and tasks](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8342–8360. Association for Computational Linguistics.
- Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. [The interplay of variant, size, and task type in Arabic pre-trained language models](#). In *Proceedings of the Sixth Arabic Natural Language Processing Workshop (WANLP)*, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.
- Mohammad M. Khader, AbdulGabbar Al-Sharafi, Mohammad Hamza Al-Sioufy, Wajdi Zaghouni, and Ali Al-Zawqari. 2024. [Munazarat 1.0: A corpus of Arabic competitive debates](#). In *Proceedings of the 6th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT) @ LREC-COLING 2024*, pages 20–30, Torino, Italia. ELRA and ICCL.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. 2024. [DSPy: Compiling declarative language model calls into self-improving pipelines](#). In *The Twelfth International Conference on Learning Representations (ICLR)*.
- J. Richard Landis and Gary G. Koch. 1977. [The measurement of observer agreement for categorical data](#). *Biometrics*, 33(1):159–174.
- John Lawrence and Chris Reed. 2019. [Argument mining: A survey](#). *Computational Linguistics*, 45(4):765–818.
- Yu-Jen Lin and Chih-Jen Lin. 2023. [On the thresholding strategy for infrequent labels in multi-label classification](#). In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM ’23)*, Birmingham, United Kingdom. ACM.
- Zachary C. Lipton, Charles Elkan, and Balakrishnan Naryanaswamy. 2014. [Thresholding classifiers to maximize F1 score](#). *Preprint*, arXiv:1402.1892. Published version: “Optimal Thresholding of Classifiers to Maximize F1 Measure”, ECML PKDD 2014.
- Llama Team, AI @ Meta. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Chiyan Loo. 2025. [Self-consistency is losing its edge: Diminishing returns and rising costs in modern LLMs](#). *Preprint*, arXiv:2511.00751.
- Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. 2021. [Long-tail learning via logit adjustment](#). In *The Ninth International Conference on Learning Representations (ICLR)*.
- Marie-Francine Moens. 2018. [Argumentation mining: How can a machine acquire common sense and world knowledge?](#) *Argument & Computation*, 9(1):1–14.
- Sara Nabhani, Nahla Bassyouni, Ali Al-Zawqari, Mohammad Khader, and Khalid Al-Khatib. 2026. Daleel 2026 shared task on Arabic argumentative discourse mining. Budapest, Hungary. Association for Computational Linguistics.
- Andreas Peldszus and Manfred Stede. 2013. [From argument diagrams to argumentation mining in texts: A survey](#). *International Journal of Cognitive Informatics and Natural Intelligence*, 7(1):1–31.
- Qwen Team. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.
- Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. 2021. [Preregistering NLP research](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. Association for Computational Linguistics.
- Andrew J. Viterbi. 1967. [Error bounds for convolutional codes and an asymptotically optimum decoding algorithm](#). *IEEE Transactions on Information Theory*, 13(2):260–269.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. [Self-consistency improves chain of thought reasoning in language models](#). In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Manuel Widmoser, Maria Leonor Pacheco, Jean Honório, and Dan Goldwasser. 2021. [Randomized deep structured prediction for discourse-level processing](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume (EACL)*. Association for Computational Linguistics.
- Ruixuan Xiao, Yiwen Dong, Junbo Zhao, Runze Wu, Minmin Lin, Gang Chen, and Haobo Wang. 2023. [FreeAL: Towards human-free active learning in the era of large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics.

Canwen Xu, Yichong Xu, Shuohang Wang, Yang Liu, Chenguang Zhu, and Julian McAuley. 2023. [Small models are valuable plug-ins for large language models](#). *Preprint*, arXiv:2305.08848.

Frances Yung, Varsha Suresh, Zaynab Reza, Mansoor Ahmad, and Vera Demberg. 2025. [Synthetic data augmentation for cross-domain implicit discourse relation recognition](#). *Preprint*, arXiv:2503.20588.

Tianyi Zhang, Linrong Cai, Jeffrey Li, Nicholas Roberts, Jinoh Lee, Frederic Sala, and Neel Guha. 2025. [Stronger than you think: Benchmarking weak supervision on realistic tasks](#). *Preprint*, arXiv:2501.07727.

Zhen Zhang, Yuhua Zhao, Hang Gao, and Mengting Hu. 2024. [LinkNER: Linking local named entity recognition models to large language models using uncertainty](#). In *Proceedings of the ACM Web Conference 2024 (WWW '24)*, Singapore. ACM.

A Closed-Track Disclosure

Per the task rules, all models, tools, prompts, and processing steps:

Models (all open-weight, $\leq 70\text{B}$, licenses verified). gemma-4-31b-it (Gemma Team, Google DeepMind, 2026): every LLM stage (Task 1 classifier, quote proposals, role decisions, ST verifier); Apache-2.0; served via OpenRouter (paid route pinned after free-route starvation); $T=0$ for reference/deploy runs, $T=0.7$ for rollouts; ChatAdapter with a deterministic retry ladder (a JSON-adapter failure mode on $\sim 2.8\%$ of atom decisions was eliminated this way; a rare whitespace-flood degeneration on $\sim 1/430$ paragraphs is contained as an empty prediction). Pretraining cut-off precedes the task’s label release, so gold-label leakage is impossible; the underlying news/debate text may predate the cutoff. qwen3-32b (Qwen Team, 2025) and llama-3.3-70b (Llama Team, AI @ Meta, 2024): one zero-shot Task 1 run each, used only as the verifier’s co-signal (and in rejected bundle experiments); $T=0$, which deviates from Qwen’s recommended stochastic sampling; Arabic is not among Llama 3’s officially supported languages: we use it only as a redundant negative signal; Llama 3.3 has no dedicated report, so we cite the herd paper, and the exact checkpoint is identified by its pinned route in the released configs. CAMELBERT-MSA-quarter (Inoue et al., 2021), Apache-2.0, pinned to an immutable revision: Task 1 paragraph classifier and Task 2 segment classifier. DSPy 3.3.0b1 (Khatab et al., 2024); GEPA compile used deepseek-v4-pro as

the reflection model (compile-time only; no task inference). No external labeled data; no dataset text in any committed artifact.

Prompts. All prompts are audit-informed zero-shot seeds with the official label definitions in a single canonical phrasing (shared between programs and metric feedback); the ST verifier and rejected CO specialist add fold-external labeled exemplars as described in the text. Verbatim prompt texts ship in the released repository (made public at camera-ready; see §1).

Evaluation-phase upload record (§6).

Stronger configurations missed the leaderboard on both tasks. Our final Task 1 package (predicted 0.7182 by the composition arithmetic) was never uploaded: the closing upload meant to carry it held an earlier probe configuration by mistake (scoring 0.711) and was never pushed to the board, so the scored Task 1 entry is probe-3 (0.712). Two closing Task 2 uploads scored 0.733 but were never pushed either, leaving probe-4 (0.7316) the scored Task 2 entry. Both are operational errors, not modeling choices, and neither is claimed as a result.

Pre/post-processing. Whitespace-only gold spans dropped (training side only); Task 1 gold re-derived from span labels with parity assertion; deterministic clause/connective atomization for role decisions; verbatim-quote alignment (exact/whitespace/normalized/fuzzy ladder) for offsets; per-label vote thresholds, encoder thresholds, CO budget, calibration weights, blend λ , and transition matrices all cross-fitted per fold on the 430-pool OOF artifacts and refit on all 430 for deployment twins; exact-duplicate span dedup after decode. Empty prediction sets are legal outputs everywhere.

B The Gate Registry

Every recorded decision, in order (Table 2); gates are composed-OOF +0.02 with $\geq 3/5$ fold wins unless stated. “Killed” = failed a pre-declared cheaper screen before the full gate.

Summary: 29 decisions, 6 adoptions, 2 external failures caught and reverted by the one-probe rule (row 13 and the OT leg of row 14), 0 families reopened. Rows 26–27 are the sharpest illustration of the margin’s purpose: every banked component reproduced its estimate almost exactly and the honest sum was still $+0.016 < +0.02$;

#	Decision (task)	Gate / screen	Result	Verdict	Registered → verdict
1	quote-then-align vs. segment (T2 program)	paired +0.02 on frozen val	0.6805 vs. 0.6542	adopt	pre-ledger
2	BFRS compile (T1)	paired +0.02	−0.042	reject	pre-ledger
3	MIPROv2 compile (T1)	paired +0.02	negative	reject	pre-ledger
4	GEPA compile (T1, 3 variants)	paired +0.02	≤ +0.003	reject	pre-ledger
5	GEPA compile (T2)	paired +0.02	+0.003	reject	pre-ledger
6	zero-shot verify stage (T1)	paired +0.02	±0.003	reject	pre-ledger
7	blind relabel stage (T2)	paired +0.02	−0.003	reject	pre-ledger
8	decision-level judge compile (T1)	optval select → frozen confirm	+0.033 → +0.001	reject	pre-ledger
9	v3 proposal-atom-role (T2)	OOF gate + frozen confirm	+0.0387; 0.6995 vs. 0.6805	adopt	pre-ledger
10	v3 Task-1 fusion	OOF gate + frozen confirm	+0.0436; frozen transfer failed	reject	pre-ledger
11	theory-variables-alone design (T1)	fitted ceiling ≥ champion	0.6417 < ~0.70	killed	e15c8f9 [†]
12	tier-1 routed composition (T1)	OOF +0.02 vs. champion-T0	+0.0225; dev 0.6642	adopt	c603bad [†]
13	mix-encoder amendment (T1)	one dev probe	0.6364 vs. 0.6642	reverted	7127672 → bcd0aca
14	routed v2 recomposition (T1)	OOF +0.02	+0.0381, 5/5; dev +0.0146; OT leg reverted → v2.1 0.6955	adopt	5eafbde → 34324ce [†]
15	CO many-shot rank feature (T1)	precision@k on ≥ 3/5 folds	0.3733 vs. 0.40 control	reject	5eafbde → f4ef8e5 [†]
16	T2 atom-vote self-consistency	fold-0 spend gate +0.005	+0.0024	killed	5eafbde → f4ef8e5 [†]
17	discourse-forest prompts (T2)	forest > flat, shuffled; +0.02	forest − shuffled = 0.0003	killed	02e19a0 → 8e3414a [†]
18	constrained relabel (T2)	OOF +0.02	+0.0182	reject	5eafbde → 1ad20a0 [†]
19	constrained relabel regate (T2)	OOF +0.02 (v2.2 constraint)	0.7133 vs. gate 0.7134	reject	9d094de
20	learned stacker (T1, screen)	beat hand rules on OOF	0.7084 vs. 0.7203	reject	922670d [†]
21	ST contrastive verifier v2 (T1)	OOF +0.02	+0.0233, 5/5; dev exact	adopt	922670d → 848c447 [†]
22	CO contrastive specialist (T1)	OOF +0.02, control-exact	0.7254 < 0.7498	reject	d713a91 → 2d02ee4
23	AN-rescue span judge (T2)	fold-0 screen +0.015	−0.0024	killed	05ac51f → 49228a5
24	S1 structural re-decode (T2)	OOF +0.02, control-exact	+0.0272, 5/5; dev +0.0400	adopt	46368f9 → fa28fa6
25	S2 learned span combiner (T2)	OOF +0.02	+0.0013, flip precision 0.467	reject	53382af → d0b1e84
26	bundle-v3 full (T1)	fixed-sequence primary	+0.0160, 4/5	reject	56d6f30 → 4607866
27	bundle-v3 ensemble-only (T1)	fixed-sequence secondary	+0.0138, 4/5	reject	56d6f30 → 4607866
28	calibration temperature τ (T2)	OOF +0.02, control-exact	−0.0003; flat in τ	reject	0f6b709 → a5c24f6
29	transductive TAPT (T1; measurement-only)	OOF +0.02, control-exact	+0.0066 (encoder-only +0.0476)	reject	cd77061 → 1e88530

Table 2: The complete adoption registry. “pre-ledger” rows predate the in-repo registration convention and are documented in the corresponding campaign reports; later rows cite the ledger and verdict commits in the released repository. [†]: design and verdict entered git together (same commit or same-day batch), so precedence over the run rests on the ledger’s contemporaneous record, not git order. Every reject/kill closed its design family permanently.

a +0.016 gain on 430 paragraphs does not earn a leaderboard probe.

C Analysis Detail

The four subsections below are the full development of the patterns summarized in §6.

C.1 Criteria verify; conventions require supervision or structure

The campaign ran four prompting attacks on convention-defined labels, all pre-declared, all failed: (i) zero-shot CO definitions (cross-fitted F_1 0.07); (ii) a CO definition judge as a fourth rank feature (0.3733 vs. 0.40 control); (iii) a CO contrastive many-shot specialist with the annotators’ own exemplars and four induced not-CO conventions in context (0.3733 again: the model argued false positives *into* CO while quoting the rules); (iv) a span-level AN-vs-AS judge (flip precision 0.44 where ≥ 0.75 was needed). Meanwhile the ST verifier (aimed at a checkable criterion) validated twice and shipped. The Webis agreement numbers explain the asymmetry (Fleiss’ κ , segment-level, pre-consolidation: AS 0.613, TE 0.591, ST 0.582, AN 0.399, OT 0.152, CO 0.114; Al-Khatib et al., 2016): CO’s convention ($\kappa=0.114$, 0.562 confusion into AS *among humans*) is statistical, not definitional; there is nothing for a verifier to check. Consistently, gemma fired CO on *every* call for

41 paragraphs whose gold is not CO, and on *no* call for 7 gold CO paragraphs: a *biased coin* no amount of voting, prompting, or calibration can flip. A dev example makes the convention concrete: the router fired CO on a debate opening that merely *frames* the motion (“this House would ban for-profit companies from using social and political causes in marketing”): gold AS/OT, because a motion is assumed for the sake of debate, not shared ground. Over-prediction of such labels is even metric-rational (Lipton et al., 2014).

C.2 Vote-scaling: why $k=5$, and why voting cannot fix ST or AN

With q_p the observed fraction of the five rollouts firing on paragraph p , Table 4 turns three design choices into consequences of vote statistics (Chen et al., 2024; Wang et al., 2023; Loo, 2025). (1) $k=5$ suffices: for the three vote-routed labels, F_1 at $k=5$ with an optimal threshold is over 99% of the infinite-vote limit ($\geq 96.9\%$ for all six); more same-prompt rollouts buy almost nothing. (2) *Voting cannot fix* ST: the sampler already finds every gold ST ($p_1=0.99$, zero inverted golds), but 26 non-gold paragraphs fire on a majority of calls: correlated false positives that more votes make *more* confident. Precision required an orthogonal signal: the verifier. (3) *Voting cannot fix* AN: half the gold AN paragraphs fire on a minor-

ity of calls ($\text{inv}^+=0.50$), capping the infinite-vote ceiling at 0.611, *below* the supervised encoder’s 0.717; routing AN to the encoder is not a preference but an inequality. Finally, deployed thresholds sit at their pooled-OOF optima (Table 3), but the interval retaining $\geq 95\%$ of peak F_1 spans 0.77 of the score range for AS against 0.016 for ST and **0.006** for CO: rare-label operating points sit on needles, and any dev $\rightarrow$ eval score shift falls off the peak: the quantitative case for CO’s count budget and ST’s verifier.

C.3 What S1 did, seen against the human matrix

Grouping the six types into coarse families (evidence TE/ST/AN, subjective CO/AS, other) localizes S1’s contribution: the within-family share of misassigned char mass fell $0.258 \rightarrow 0.182$ while family-level F_1 barely moved ($0.7512 \rightarrow 0.7541$): S1’s gain is within-family disambiguation. The residual concentrates in the one cross-family confusion the decoder cannot reach, gold-AN mass labeled AS (0.330): on dev, the gold AN span “the reaffirmation to activate the Loss and Damage Fund” (a settled event, hence anecdote by convention) was absorbed into a larger AS span. The same confusion leads the human matrix (0.277; Al-Khatib et al., 2016), and our 6-label OOF (0.7206) against the 3-family score (0.7541) bounds what finer typing can still add (§2).

Per-label test decomposition (§6). Task 1 test F_1 : AS 0.9408, ST 0.9286 (recall 1.0, so the verifier’s zero drops were correct), OT 0.7805, TE 0.6724 (editorial recall 0.258 vs. 0.795 debate), AN 0.4628 (76 fired against 45 golds), CO 0.1143 (the frozen budget fired 25 paragraphs against 10 test golds, the recorded overfiring risk realized). Editorial OT scored 0.0 on 6 supports, the blind spot dev had already exposed.

C.4 Negative results

Every rejected family and its gate is in Appendix B; the sharpest three: the **cross-model vote bundle** delivered every banked component’s estimate almost exactly and the honest sum was still $+0.0160 < +0.02$: closed without a probe, exactly the case the winner’s-curse guard exists for; a **shuffled-graph placebo** scored within 0.0003 of the true discourse forest: the structured-prompt gain was graph-shape, not attachment; and **transductive TAPT** (Gururangan et al., 2020) lifted

the raw encoder $+0.0476$ OOF but the composed system only $+0.0066$: textbook composition dilution (the encoder feeds only AN/OT and CO rank features). Also closed at their gates: prompt optimizers on monolithic signatures (-0.04 to $+0.003$; the only win came after decomposition to single decisions), a decision-level judge compile ($+0.033$ selection, $+0.001$ confirmation), learned stacking, learned span arbitration, calibration temperature, and synthetic-label routes (Xu et al., 2023; Fu et al., 2021; Zhang et al., 2024; Yung et al., 2025; Zhang et al., 2025; Di Palo et al., 2024; Xiao et al., 2023).

Threshold audit (§C.2). Deployed encoder thresholds vs. pooled-OOF optima, the $F^*/2$ plug-in point, and the width of the $\geq 95\%$ -of-peak interval:

	θ_{dep}	θ^*	F^*	$F^*/2$	95%-width
CO	.181	.178	.350	.175	.006
AS	.473	.471	.916	.458	.774
TE	.288	.286	.661	.331	.158
ST	.164	.164	.326	.163	.016
AN	.384	.384	.717	.359	.288
OT	.484	.482	.814	.407	.376

Table 3: Every deployed threshold is at its pooled optimum and near the calibrated plug-in rule; what differs across labels by two orders of magnitude is the width of the safe operating region.

Vote thresholds. Deployed integer vote thresholds $(\theta_{\text{AS}}, \theta_{\text{TE}}, \theta_{\text{ST}}) = (2, 4, 4)$ are exactly the pooled-OOF optima; the full F_1 -by- θ and F_1 -by- k curves (including the fixed-majority inverse-U on OT) accompany the released analysis artifacts. Table 4 gives the per-label vote model behind §C.2: with q_p the observed vote fraction of paragraph p , expected F_1 at k votes uses $E[\text{TP}] = \sum_{\text{gold}} P(\text{Binom}(k, q_p) \geq \theta)$ (and $E[\text{FP}]$ likewise) at the optimal integer θ ; the $k \rightarrow \infty$ limit fires p iff $q_p > \tau$ at the best observed τ ; inv^+ is the share of gold paragraphs with $q_p < 0.5$, unreachable by any majority.

Transfer ledger (§6). What each development submission was predicted to score externally, by which estimator, and what it scored:

Progression and official scores (§6). Development-phase progression and official evaluation-phase scores:

Confusion mass (§C.3). Row-normalized gold char-mass destinations for the final S1 output, selected rows: gold-CO mass goes 0.563 to CO,

	α	p_1	p_2	inv^+	$F_1@5$	$F_1@∞$
CO	.058	.63	.16	.36	.327	.337
AS	.756	.94	.35	.06	.920	.922
TE	.258	.86	.13	.13	.791	.793
ST	.044	.99	.07	.00	.609	.613
AN	.300	.49	.10	.50	.603	.611
OT	.372	.84	.25	.16	.746	.748

Table 4: Per-label vote model from the five rollouts: prevalence α , per-call fire rate on gold (p_1) and non-gold (p_2) paragraphs, fraction of gold items fired on $<50\%$ of calls (inv^+ : unreachable by any majority), and expected F_1 at $k=5$ vs. the infinite-vote limit, both at the optimal integer threshold.

Submission	Predicted	Observed	Δ
champion (val 182)	0.7024	0.6159	-0.087
routed v1 (OOF)	0.6721	0.6642	-0.008
mix amendment (OOF)	0.6955	0.6364	-0.059
routed v2 (OOF)	0.7102	0.6788	-0.031
v2.1 (arithmetic)	0.6955	0.6955	± 0
v2.2 (arithmetic)	0.7089	0.7089	± 0
v3 (frozen 182)	0.6995	0.6847	-0.015
S1 (arithmetic+OOF)	≈ 0.712	0.7247	+0.013

Table 5: The transfer ledger. Single-shot readouts on 182 paragraphs (rows 1, 7) carry ± 0.03 – 0.04 provider noise; cross-fitted composed OOF transfers between -0.008 and -0.059 ; exact composition arithmetic transfers exactly.

0.286 to AS; gold-AN mass goes 0.382 to AN, 0.330 to AS, 0.112 to TE, with only 0.037 uncovered by any predicted span: the AN deficit is typed-coverage, not extraction-coverage, and it is the one large cross-family residual. Full 6×7 matrices (before/after S1, including uncovered mass) are in the released analysis artifacts.

D Reproducibility and Cost

Every run writes a hash-bound artifact set (exact config, data hashes, fold manifest, metrics, provenance, completion marker); adoption reports verify fold membership by hash; deployment twins reuse the immutable model/threshold contracts of their OOF runs. Split seeds: 20260709 for the stratified 430/182 selection/confirmation split, 20260710 for the 5-fold paragraph-grouped, genre-stratified partition of the 430. Encoders are pinned to 40-hex revisions and trained fp32 on CPU for bit-stable deployment parity (Task 1: six-sigmoid paragraph classifier, 4 epochs, lr $2 \cdot 10^{-5}$, 512-token windows with stride-128 overflow, max pooling; Task 2: seven-way segment classifier, same recipe); fp16 GPU replication of the primary encoder run reproduced digit-for-digit across duplicated jobs. LLM runs pin provider, model route, temperature, adapter, and date; the DSPy disk cache makes every reported number re-

System (chronological)	T1 dev	T2 dev
zero-shot champion	0.6159	—
v3 pipeline	—	0.6847
+ mix-encoder amendment [†]	0.6364	
routed v2.1 (OT revert)	0.6955	
routed v2.2 (+ST verifier)	0.7089	
S1 structural re-decode		0.7247
organizers’ baseline	0.358	—
evaluation, frozen recipes (test, 213)	0.6499	0.7055
editorial only	0.4595	0.6195
debate only	0.7112	0.7364
evaluation, final probes	0.712	0.7316
final rank (teams)	1/10	5/8

Table 6: Progression and official scores (closed track, *both* setting; intermediate routed v1 0.6642 and v2 0.6788 omitted for space). [†]Failed external check, reverted same day. The evaluation rows report (i) the pinned recipes v2.2 and S1 run once, unchanged, on the test input, and (ii) the final probe submissions of the disclosed evaluation-week campaign (§6).

derivable without new API calls. Splits, fold seeds, and threshold-fitting code are released; the official scorers are used through parity-tested ports. Approximate paid-API ledger: model bake-off \$3.8; optimizer campaigns $\approx \$4$; v3 role campaign $\approx \$13.5$; routing/rollout evidence $\approx \$13$; verifier, specialist, and rescue gates $\approx \$6$; development probes and deploy twins $\approx \$6$; total $\approx \text{US\$}50$. Encoder training: ≈ 30 Kaggle GPU-hours (screens and multi-seed evidence) plus CPU runs; the one TAPT measurement added ≈ 12 CPU-hours. Evaluation-phase inference (2026-07-27) was one pass per pinned stage over the 213-paragraph test input: quote extraction, role decisions on 1,912 atoms, five vote rollouts, two co-signal runs, the seed judge (16 calls) and verifier (45 calls), $\approx 3.9\text{k}$ LLM calls in total, for $\approx \text{US\$}10$, plus ≈ 35 CPU-minutes for the two encoder deploys; every stage completed error-free on the first attempt, and the two deploy-twin controls reproduced their development outputs byte-identically before any test inference ran.